

My first document

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1 Section

Hello World!

1.1 Subsetion

Stucturing a document is easy!

1.1.1 Subsubsection

More Text.

Paragraph some more text.

Subparagraph Even more text.

2 Another section

to type an equation

$$f(x) = x^2 \tag{1}$$

to remove the automatic numbering be using amsmath package

$$f(x) = x^2$$

To make use of the inline math feature. This formula $f(x) = x^2$ is an example. The equation and align environment

$$\begin{aligned} 1 + 2 &= 3 \\ 1 &= 3 - 2 \end{aligned}$$

Fractions and more

$$\begin{aligned} f(x) &= x^2 \\ g(x) &= \frac{1}{x} \\ F1(x) &= \int_b^a \frac{1}{3} x^3 \\ F2(x) &= \int_{min}^{max} \frac{1}{3} x^3 \end{aligned}$$

$$\begin{aligned} F3(x) &= \left(\frac{1}{\sqrt{x}}\right) \\ F4(x) &= \left(\frac{1}{\sqrt{x}}\right) \\ F5(x) &= \left[\frac{1}{\sqrt{x}}\right] \\ F6(x) &= \left|\frac{1}{\sqrt{x}}\right| \end{aligned}$$

Matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

For the inline $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

to use brackets you have to ways
 use name documents - for $\begin{bmatrix}$ use bmatrix - for $\begin{pmatrix}$ use pmatrix - for $\det$ use
 vmatrix

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$\begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix}$$

$$\begin{vmatrix} a & b \\ b & a \end{vmatrix}$$

$$\det1 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$\det2 \begin{vmatrix} a & b \\ b & a \end{vmatrix}$$

$$\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Miscellaneous functions $\log x$, $\log_a x$, $\sqrt{x}$, $\sqrt[n]{x}$, $\frac{u(x)}{v(x)}$
 Trigonometric functions $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, $\csc x$.
 Integrals $\int f(x)dx$, $\int_a^b f(x)dx$, $\int_D f(x)$ also you can continue like this :

$$\iint f(x,y)dx dy$$

$$\iiint f(x,y,z)dx dy dz$$

$$\oint_C F ds$$



Figure 1: A syria.

Captioned images / figures in LaTeX

Figure 1 shows a what...